

STUDY MATERIAL

SSC CGL Study Material

Demo Edition - Sample Chapters

Quantitative Aptitude - Reasoning - English - General Awareness

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Table of Contents

About this demo	3
Chapter 1 - Quantitative Aptitude: Number System	4
1.1 Number System - Theory	4
1.2 Important Formulas	5
1.3 Worked Examples	6
Chapter 2 - Quantitative Aptitude: Percentage	7
2.1 Percentage - Concepts	7
2.2 Short-cut Techniques	8
2.3 Practice Set 1	9
Chapter 3 - General Intelligence & Reasoning	10
3.1 Series Completion	10
3.2 Analogy	11
Chapter 4 - English Comprehension	12
4.1 Common Error Patterns	12
4.2 Reading Comprehension Tips	13
Chapter 5 - General Awareness Quick Notes	14
Practice Question Bank (with answers)	15
About StudyElites & How to Get Full Access	17

About this demo

This is a **free demo PDF** from StudyElites. It contains sample chapters from our SSC CGL Study Material so you can preview the structure, depth and quality of the full product before purchasing.

The full version (delivered after purchase via Razorpay) contains the complete chapter set, additional practice sets, previous-year question walkthroughs, and topic-wise quick-revision sheets for SSC CGL Tier I and Tier II. This demo covers approximately 15-20% of the full material.

What's inside this demo

- Chapter 1 - Number System (theory + formulas + worked examples)
- Chapter 2 - Percentage (concepts + short-cuts + practice set)
- Chapter 3 - Reasoning (series + analogy samples)
- Chapter 4 - English comprehension (error patterns + RC tips)
- Chapter 5 - General Awareness quick notes
- 10-question practice bank with answer key

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Chapter 1

Quantitative Aptitude - Number System

1.1 Number System - Theory

The number system is the foundation of quantitative aptitude. A clear understanding of how numbers are classified, how divisibility works, and how to manipulate fractions, decimals and surds is essential before attempting any SSC CGL Quant question. This chapter walks through the classification of numbers, the key divisibility tests, and the relationship between fractions, decimals and percentages - concepts that recur in almost every Tier I paper.

Numbers are first classified into **natural numbers** (1, 2, 3, ...), **whole numbers** (0, 1, 2, ...), **integers** (... , -2, -1, 0, 1, 2, ...), and **rational numbers** (numbers expressible as p/q where q is not 0). Numbers that cannot be expressed as p/q - such as $\sqrt{2}$, $\sqrt{3}$ and π - are called **irrational numbers**. Together, rational and irrational numbers form the set of **real numbers**.

For SSC CGL, the most practically useful sub-topic within the number system is **divisibility**. Knowing the divisibility rules for 2, 3, 4, 5, 6, 8, 9, 10, 11 and 12 lets you eliminate options quickly without performing full division. For example, a number is divisible by 11 if the alternating sum of its digits is 0 or a multiple of 11 - useful when checking whether a 5- or 6-digit number is divisible by 11 under time pressure.

1.2 Important Formulas

Divisibility rule of 11: |sum of digits at odd positions - sum of digits at even positions| is divisible by 11

HCF x LCM = Product of two numbers (for two positive integers)

$$(a + b)^2 = a^2 + 2ab + b^2 \quad (a - b)^2 = a^2 - 2ab + b^2$$

$$a^2 - b^2 = (a + b)(a - b) \quad (a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2) \quad a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

1.3 Worked Examples

Example 1. Find the unit digit of 7^{2025} .

The unit digit of powers of 7 follows a cycle of length 4: 7, 9, 3, 1, 7, 9, 3, 1, ... So we divide the exponent by 4 and look at the remainder. $2025 / 4$ gives remainder 1, so the unit digit equals the first element of the cycle, which is 7.

Example 2. Find the HCF of 96 and 408 using the division method.

Apply Euclid's algorithm: $408 = 96 \times 4 + 24$; $96 = 24 \times 4 + 0$. The last non-zero remainder is 24, so $\text{HCF}(96, 408) = 24$. Cross-check: $\text{LCM} = (96 \times 408) / 24 = 1632$.

Example 3. Simplify: $(\sqrt{5} + \sqrt{3})(\sqrt{5} - \sqrt{3})$.

Using the identity $(a + b)(a - b) = a^2 - b^2$ with $a = \sqrt{5}$ and $b = \sqrt{3}$: $(\sqrt{5})^2 - (\sqrt{3})^2 = 5 - 3 = 2$.

Short-cut reminder

When asked for the unit digit of a^n , write out the first 4 powers of a 's unit digit, observe the cycle, then reduce $n \bmod 4$. This saves 30-60 seconds per question in Tier I.

Chapter 2

Quantitative Aptitude - Percentage

2.1 Percentage - Concepts

Percentage is one of the highest-yield topics in SSC CGL Quant. Questions on percentage, profit-loss, simple/compound interest and successive percentage changes all reduce to the same underlying idea: expressing one quantity as a fraction of another with denominator 100. Mastering percentage shortcuts can save 4-6 minutes in the Tier I paper, which is often the difference between selection and missing the cut-off.

A percentage is a fraction with denominator 100. The statement "x is 25% of y" means $x = (25/100) \times y$. To convert a fraction to a percentage, multiply by 100 and add the % sign; to convert a percentage to a fraction, divide by 100 and simplify.

2.2 Short-cut Techniques

Successive percentage change formula: if A changes by x% then by y%, net change = $x + y + (xy / 100)$

If a number is increased by x%, the new number = original $\times (1 + x/100)$; if decreased by x%, new = original $\times (1 - x/100)$

If price increases by x%, consumption must decrease by $(100x / (100 + x))\%$ to keep expenditure constant

Example. A shopkeeper marks his goods 40% above cost price and gives a 20% discount. What is his profit percentage?

Let cost price = 100. Marked price = 140. Discount = 20% of 140 = 28. Selling price = 140 - 28 = 112. Profit = 12, so profit % = **12%**. Using the successive-change shortcut: +40 then -20 gives $40 - 20 - (40 \times 20 / 100) = 20 - 8 = \mathbf{12\%}$. Same answer, half the work.

2.3 Practice Set 1

Q1. A student scored 360 marks out of 500. What is his percentage?

- (A) 70%
- (B) 72%
- (C) 75%
- (D) 80%

Answer: (B) 72% - $(360/500) \times 100 = 72$.

Q2. The price of a pen increases from Rs.40 to Rs.50. What is the percentage increase?

- (A) 20%
- (B) 25%
- (C) 30%
- (D) 35%

Answer: (B) 25% - increase = 10; $(10/40) \times 100 = 25\%$.

Q3. If A's salary is 20% more than B's, by what percentage is B's salary less than A's?

- (A) 16.67%
- (B) 20%
- (C) 25%
- (D) 33.33%

Answer: (A) 16.67% - required % = $(20/120) \times 100 = 16.67\%$.

Chapter 3

General Intelligence & Reasoning

3.1 Series Completion

Series questions give you a sequence of numbers or letters and ask you to identify the next term. The trick is to first check whether the series is increasing, decreasing, or alternating, then test common patterns: arithmetic progression, geometric progression, differences of differences, squares/cubes, prime numbers, or a combination of two interleaved series.

Example. Find the next term: 2, 6, 12, 20, 30, ?

Differences: $6-2 = 4$, $12-6 = 6$, $20-12 = 8$, $30-20 = 10$. The differences themselves increase by 2 each step, so the next difference is 12, giving $30 + 12 = 42$. (Alternative pattern: $n^2 + n$, where $n = 1, 2, 3, 4, 5, 6$ produces 2, 6, 12, 20, 30, 42.)

3.2 Analogy

Analogy questions present a pair (A : B) and ask which of the given options completes a second pair (C : ?) in the same way. The key is to identify the underlying relationship - synonym/antonym, part-whole, cause-effect, worker-tool, etc. - then match it to the second pair.

Example. Doctor : Hospital :: Teacher : ?

The relationship is "worker : workplace". A doctor works in a hospital; a teacher works in a **school**.

Example. 4 : 16 :: 5 : ?

The relationship is "number : square of number". $4^2 = 16$, so $5^2 = 25$.

Chapter 4

English Comprehension

4.1 Common Error Patterns

SSC CGL's "spot the error" questions follow predictable patterns. Most errors fall into one of these categories: subject-verb agreement, article usage, tense consistency, preposition errors, pronoun-antecedent agreement, and confusion between similar-sounding words (e.g., affect/effect, advice/advise, stationary/stationery). Practising 100-150 error-spotting questions builds an intuition that lets you spot the wrong part within seconds.

Example. Identify the error: *Neither of the boys have completed their homework.*

Error: "have" and "their". "Neither" is singular and takes a singular verb and singular pronoun: *Neither of the boys has completed his homework.*

4.2 Reading Comprehension Tips

For SSC CGL RC passages, skimming the questions *before* reading the passage lets you scan for specific keywords instead of reading every word. Most Tier I RC questions are direct - they test whether you can locate a specific fact in the passage, not whether you can analyse tone. Aim to spend no more than 5-6 minutes per RC passage of 250-350 words.

Quick RC strategy

1. Read the first sentence of the passage carefully (sets context).
2. Skim questions to identify keywords.
3. Scan the passage for those keywords; mark the line.
4. Answer direct questions first, leave inference questions for last.
5. Never re-read the whole passage - go back only to marked lines.

Chapter 5

General Awareness - Quick Notes

The SSC CGL General Awareness section covers Static GK (Polity, History, Geography, Economy), Current Affairs of the last 6-8 months, and General Science (Physics, Chemistry, Biology). Below are a few high-yield facts. The full version of this PDF contains 200+ such one-liner facts organised by topic.

Polity - Key Facts

- The Constituent Assembly of India first met on **9 December 1946**.
- The Constitution of India came into effect on **26 January 1950**.
- The 42nd Amendment (1976) added the words "Socialist", "Secular" and "Integrity" to the Preamble.
- The President of India is elected by an **Electoral College** comprising elected members of both Houses of Parliament and the Legislative Assemblies.
- The maximum strength of the Lok Sabha is **552** (530 from states + 20 from Union Territories + 2 Anglo-Indians, though the latter is now abolished).

History - Key Facts

- The Quit India Movement was launched on **8 August 1942** at the Bombay session of the Indian National Congress.
- The first battle of Panipat (1526) marked the beginning of the **Mughal Empire** in India.
- The Permanent Settlement (1793) was introduced by **Lord Cornwallis** in Bengal and Bihar.
- The Indian National Congress was founded in **1885** by A. O. Hume.

Geography - Key Facts

- The Tropic of Cancer (23 deg 26 min N) passes through **8 Indian states**: Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura and Mizoram.
- The longest river in India is the **Ganga** (2,525 km).
- The highest peak in India is **Kanchenjunga** (8,586 m), part of the Himalayan range on the Sikkim-Nepal border.
- The Andaman and Nicobar Islands lie in the **Bay of Bengal**.

Practice Question Bank

10 sample questions with answer key - covers all four sections.

Quantitative Aptitude

Q1. Find the LCM of 12, 18 and 30.

- (A) 60
- (B) 90
- (C) 180
- (D) 360

Answer: (C) 180 - $12 = 2^2 \times 3$, $18 = 2 \times 3^2$, $30 = 2 \times 3 \times 5$; $LCM = 2^2 \times 3^2 \times 5 = 180$.

Q2. A man buys an article for Rs.600 and sells it at a profit of 25%. What is the selling price?

- (A) Rs.700
- (B) Rs.720
- (C) Rs.750
- (D) Rs.800

Answer: (C) Rs.750 - $SP = 600 \times (1 + 25/100) = 600 \times 1.25 = 750$.

Q3. If the simple interest on Rs.5000 for 2 years is Rs.800, what is the rate?

- (A) 6%
- (B) 7%
- (C) 8%
- (D) 9%

Answer: (C) 8% - $800 = (5000 \times R \times 2)/100$, so $R = (800 \times 100)/(5000 \times 2) = 8$.

General Intelligence & Reasoning

Q4. Complete the series: 1, 4, 9, 16, 25, ?

- (A) 30
- (B) 36
- (C) 49
- (D) 64

Answer: (B) 36 - squares of natural numbers: $6^2 = 36$.

Q5. If FRIEND is coded as GSJFOE, how is HUSBAND coded?

- (A) ITTCBOE
- (B) IVTCBOE
- (C) ITUCBOE
- (D) IVUCBOE

Answer: (A) ITTCBOE - each letter shifted +1.

English Comprehension

Q6. Choose the correct synonym of *ABUNDANT*:

- (A) Scarce
- (B) Plentiful
- (C) Limited
- (D) Few

Answer: (B) Plentiful.

Q7. Identify the error: *He is one of the most intelligent boy in the class.*

- (A) He is
- (B) one of the most intelligent boy
- (C) in the class
- (D) No error

Answer: (B) - should be "one of the most intelligent boys".

General Awareness

Q8. Who is known as the "Father of the Indian Constitution"?

- (A) Jawaharlal Nehru
- (B) B. R. Ambedkar
- (C) Rajendra Prasad
- (D) Sardar Patel

Answer: (B) B. R. Ambedkar.

Q9. The Tropic of Cancer does NOT pass through which of these states?

- (A) Rajasthan
- (B) Odisha
- (C) Tripura
- (D) Mizoram

Answer: (B) Odisha.

Q10. The Reserve Bank of India was established in which year?

- (A) 1935
- (B) 1947
- (C) 1950
- (D) 1969

Answer: (A) 1935 - RBI was established on 1 April 1935.

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